

Part D: Outlier Handling

Z-Score Method

```
customer_id age gender region education_level employment_type \
0 1 59 Female East High School Unemployed
1 2 49 Male West Post-Graduate Salaried
2 3 35 Male East Graduate Salaried
3 4 28 Female West Graduate Salaried
4 5 41 Male South High School Salaried
.. ... ..
995 996 58 Male East Post-Graduate Self-Employed
996 997 22 Female South Post-Graduate Unemployed
997 998 27 Male East Post-Graduate Salaried
998 999 38 Male South High School Unemployed
999 1000 47 Male South Post-Graduate Unemployed

annual_income loan_amount loan_purpose credit_score repayment_history \
0 731147.0 427750.4 Business 815.0 8
1 1301267.0 269510.0 Business 696.0 0
2 758988.0 462625.4 Education 405.0 4
3 296393.0 292551.2 Car 384.0 2
4 1445422.0 516671.4 Car 513.0 5
.. ... ..
995 1276778.0 485840.0 Home 842.0 3
996 639358.0 171897.0 Home 531.0 4
997 1048124.0 570082.0 Education 840.0 9
998 937744.0 349488.0 Car 494.0 4
999 220855.0 133791.0 Education 805.0 0

transaction_count spending_ratio join_date default_flag \
0 95 54.536741 2020-01-01 0
1 162 69.063641 2020-01-02 1
2 11 36.848740 2020-01-03 0
3 351 71.208527 2020-01-04 0
4 398 44.621367 2020-01-05 0
.. ... ..
995 333 64.870102 2022-09-22 1
996 393 76.661703 2022-09-23 0
997 72 29.996718 2022-09-24 0
998 233 58.762281 2022-09-25 1
999 234 56.815583 2022-09-26 1

income_missing
0 1
1 1
2 1
3 1
4 1
.. ...
995 0
996 0
997 0
998 0
999 0

[1000 rows x 16 columns]
```

IQR Method

```
Before Winsorization
count    1.000000e+03
mean     8.507284e+05
std      3.579352e+05
min      2.781160e+05
25%      5.358800e+05
50%      8.488240e+05
75%      1.142660e+06
max      1.431266e+06
Name: annual_income, dtype: float64
```

```
After Winsorization
count    1.000000e+03
mean     8.507284e+05
std      3.579352e+05
min      2.781160e+05
25%      5.358800e+05
50%      8.488240e+05
75%      1.142660e+06
max      1.431266e+06
Name: annual_income, dtype: float64
```

Percentile Method

```
1st Percentile : 278116.0
99th Percentile : 1431266.0
```

```
Original Shape : (1000, 16)
After Percentile Method : (1000, 16)
```
